

Linear Regression: Foundations, Fitting, and Evaluation

From a single predictor to iterative optimization

Rodrigo Gonzalez, Ph.D.

1.

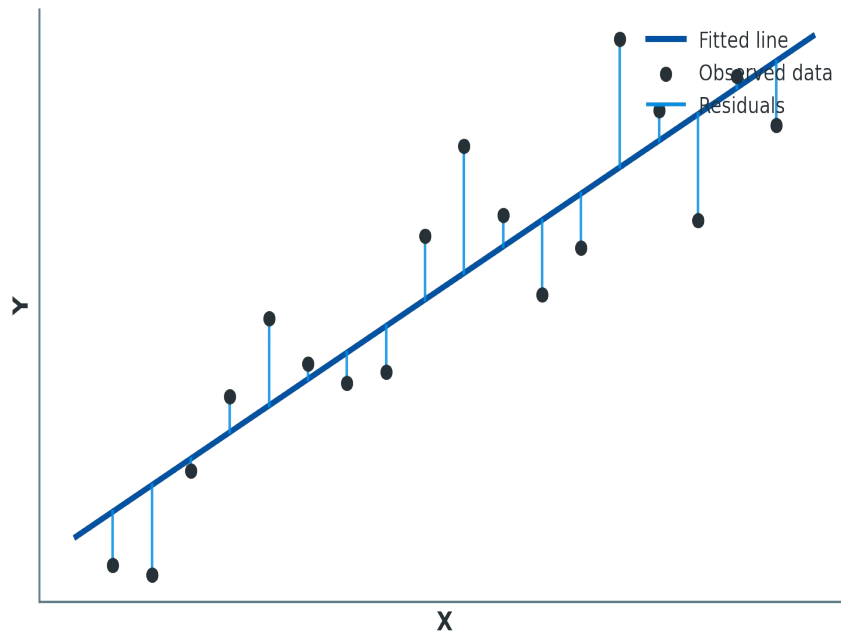
Part 1

Univariable Linear Regression

The Simple Linear Regression Model

- Models the relationship between one predictor X and response Y
- Assumes a straight-line relationship, plus random error
- Fitting means choosing β_0 and β_1 that best trace the data
- Residuals are vertical distances between points and the line

$$y = \beta_0 + \beta_1 x + \varepsilon$$



Scatter of X vs. Y with best-fit line and residuals

Interpreting Slope and Intercept

- β_0 (intercept): predicted Y when $X = 0$
- β_1 (slope): expected change in Y per one-unit increase in X
- Sign of β_1 shows direction of the relationship
- Magnitude of β_1 shows strength, in the original units of Y
- Extrapolating far outside the observed X range is risky

Geometric intuition: the least-squares line minimizes the total squared vertical distance from every point

2.

Part 2

Multivariable Linear Regression

The Multiple Regression Model

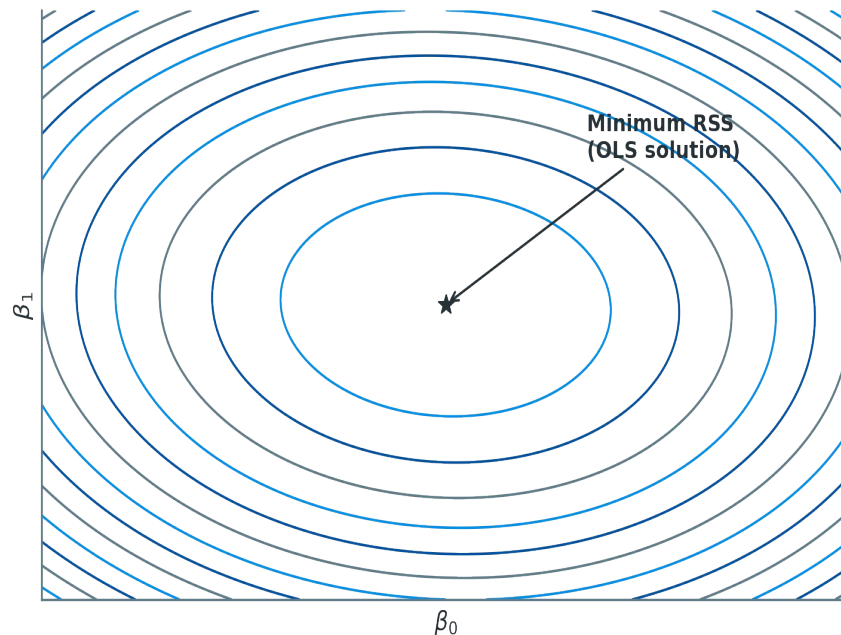
- Extends the model to p predictors $X_1, X_2, \dots, X_p$
- Each β_j is the effect of X_j holding all other predictors fixed
- Compactly written in matrix form for estimation and computation
- n observations, p predictors $\rightarrow X$ is an $n \times (p+1)$ matrix

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$$

$$\text{Matrix form: } y = X\beta + \varepsilon$$

$$\text{RSS} = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

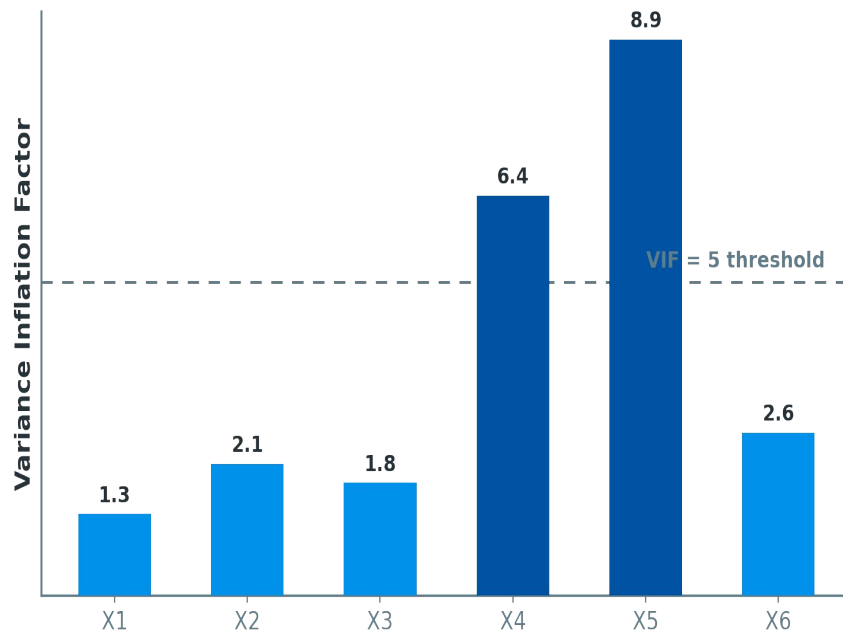
$$\text{RSS} = e_1^2 + e_2^2 + \dots + e_n^2,$$



Residual Sum of Squares (RSS) surface over (β_0, β_1) — the fitting problem in 2D

Multicollinearity, Interactions, and Dummies

- Multicollinearity: predictors correlated with each other
- Detected with the Variance Inflation Factor (VIF); $VIF > 5$ is a concern
- Interaction terms let one predictor's effect depend on another
- Categorical predictors enter via dummy (one-hot) variables



VIF per predictor, flagged against a common threshold

3.

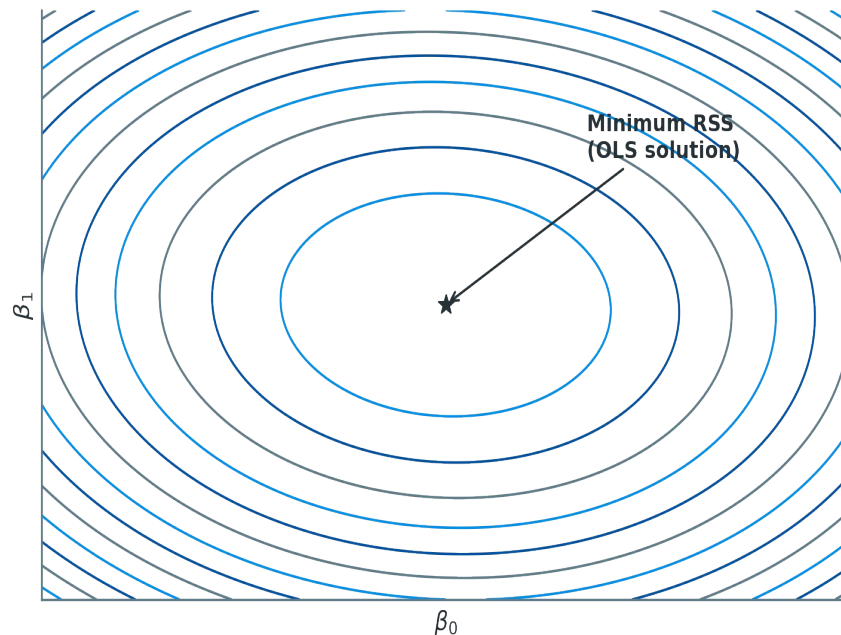
Part 3

OLS Adjustment (Closed-Form)

Minimizing the Residual Sum of Squares

- Ordinary Least Squares picks β that minimizes total squared error
- The RSS surface over β is a convex bowl with one minimum
- Setting the gradient to zero gives a closed-form solution
- The Normal Equations solve for β^* directly, no iteration needed

$$\beta^* = (X^T X)^{-1} X^T y$$



Plot: RSS contour bowl with the OLS minimum marked

The Gauss-Markov Assumptions

Linearity

The relationship between X and Y is linear in the parameters

Independence

Observations (and their errors) are independent of each other

Homoscedasticity

Error variance is constant across all levels of X

No multicollinearity

Predictors are not perfectly linearly related

Normality (for inference)

Errors are normally distributed, needed for valid p-values

When the Closed-Form Breaks Down

- $(X^T X)^{-1}$ requires $X^T X$ to be invertible (non-singular)
- Breaks down when p is large relative to n , or predictors are collinear
- Computing a large matrix inverse is expensive: $O(p^3)$
- This motivates iterative methods that scale to large, complex data

Next: fitting β iteratively with gradient descent

4.

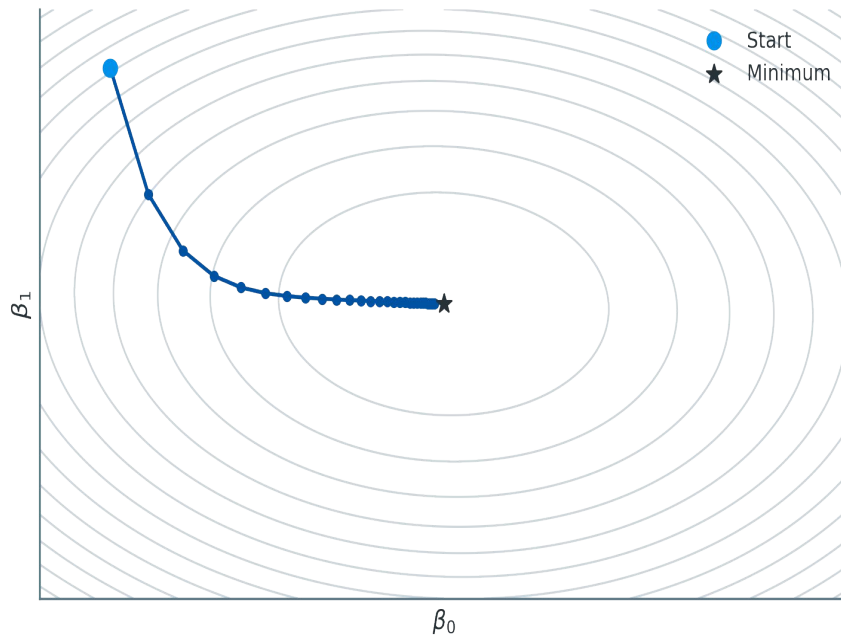
Part 4

Iterative Adjustment: Gradient Descent

Cost Function and Gradient Descent

- Cost function $J(\beta)$ is the mean squared error over training data
- Gradient descent moves β opposite the gradient of $J(\beta)$
- Also called steepest descent: the direction of fastest decrease
- Learning rate α controls the size of each update step

$$\beta \leftarrow \beta - \alpha \nabla J(\beta)$$



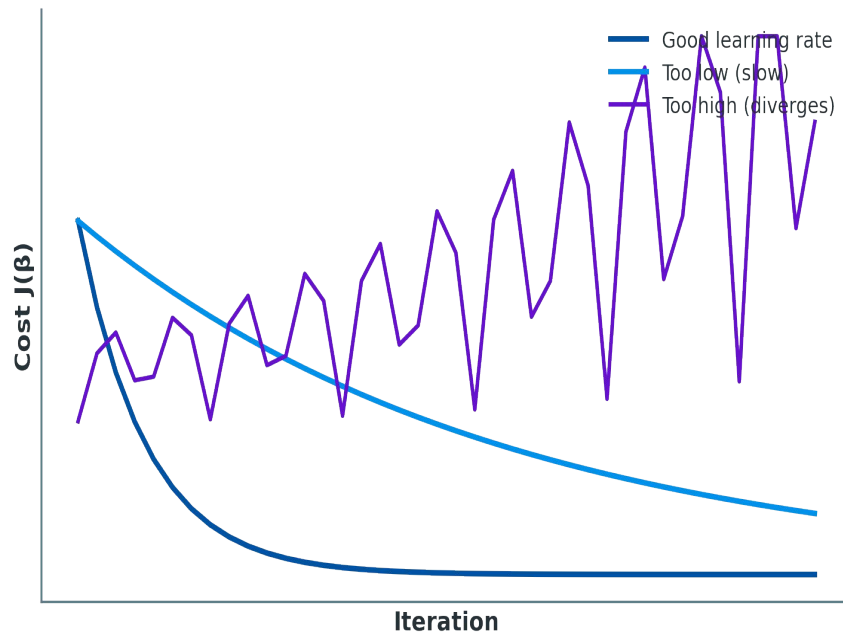
Plot: Gradient descent path converging on the RSS contour

Batch vs. Stochastic vs. Mini-Batch

	Batch GD	Stochastic GD	Mini-Batch GD
Data per update	Entire training set	One random example	Small random subset
Update stability	Smooth, stable path	Noisy, erratic path	Balanced
Speed per step	Slow	Very fast	Fast
Typical use	Small datasets	Streaming / huge data	Most common in practice

Learning Rate and Convergence

- Too small α : convergence is very slow, many iterations needed
- Too large α : updates overshoot and the cost can diverge
- A well-tuned α converges smoothly to the minimum
- Feature scaling (standardizing X) speeds up and stabilizes convergence



Plot: Cost vs. iteration for low, good, and high learning rates

5.

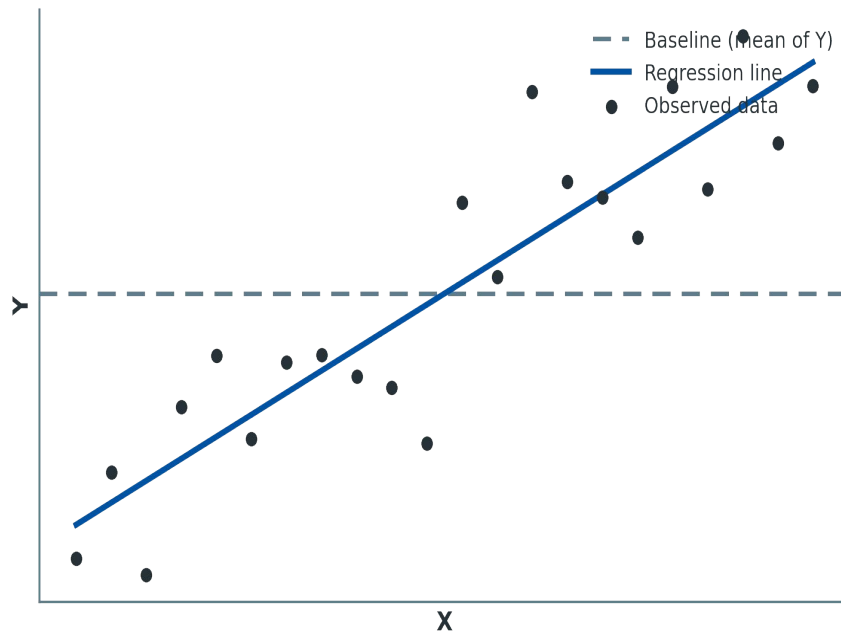
Part 5

Key Metrics for Linear Regression

R^2 , Adjusted R^2 , and Error Metrics

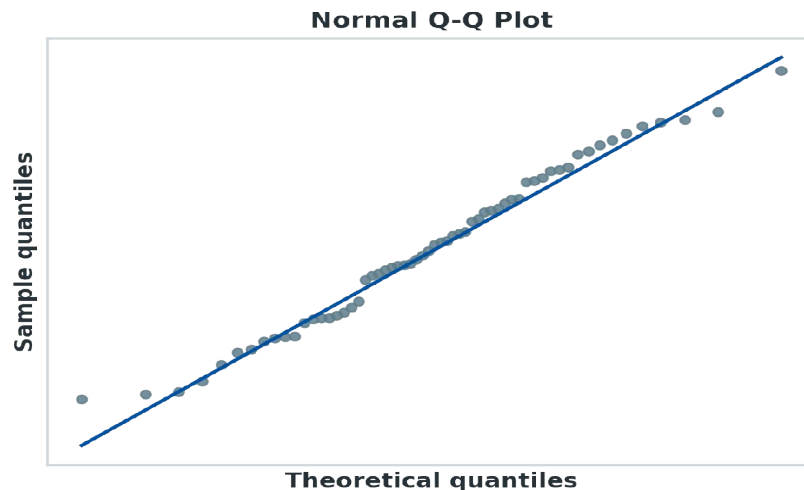
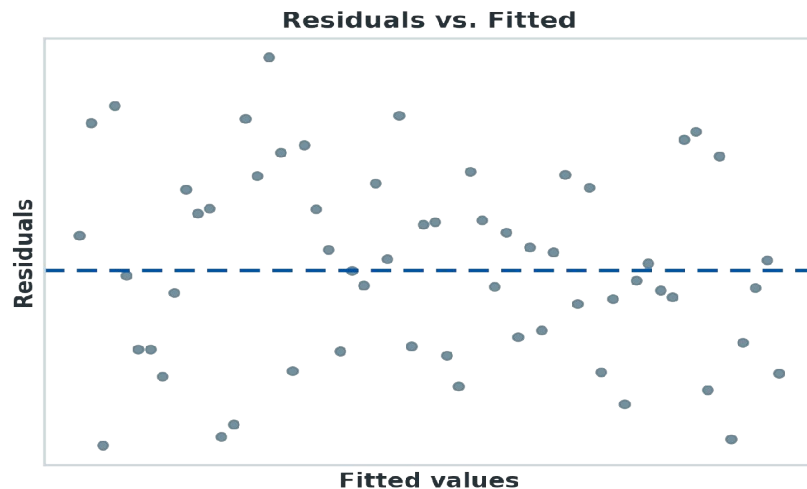
- R^2 : share of variance in Y explained by the model, vs. the mean baseline
- Adjusted R^2 penalizes adding predictors that don't help
- MSE / RMSE: average squared / root-mean-squared error
- MAE: average absolute error, more robust to outliers

$$R^2 = 1 - \text{RSS}/\text{TSS} \quad \text{RMSE} = \sqrt{(\text{RSS}/n)}$$



Plot: Regression line vs. mean baseline — the basis of R^2

Significance and Residual Diagnostics



- F-statistic: tests whether the model, as a whole, is significant
- Coefficient p-values / CIs: test each β_j individually against zero
- Residuals vs. fitted: should show no pattern (checks linearity, variance)
- Q-Q plot: points on the diagonal indicate normally distributed residuals

Thanks!

Any questions?